Yiruo Xu

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Postdoctoral Fellow

RESEARCH INTERESTS

Geochemistry, Metamorphic petrology, Tectonics, Cosmochemistry.

EDUCATION

2021-2026	University of Michigan, Ann Arbor, MI, USA **PhD in Earth and Environmental Sciences**: emphasis in petrology and high-temperature geochemistry
2017-2021	Vanderbilt University, Nashville, TN, USA **Bachelor of Arts, Summa Cum Laude, w/ Highest Honors in Earth and Environmental Sciences** Bachelor of Arts in Economics

PUBLICATIONS

2026	**Xu, Y.**, Holder, R. M., Hames, W. E. (2026). Rapid metamorphic cooling of the Quetico Subprovince, Canada, challenges the paradigm of long-lived slow-cooling Archean metamorphism. *Precambrian Research*. In review.
2026	Neumann, S., Holder, R., Connop, C., Wang, J. W., **Xu, Y.**, Neill, O., Kapp, P., Worthington, J., Hernández-Uribe, D., He, J. (2026). Intrinsic Challenges in the Interpretation of Zirconium-in-Titanite Thermometry and Quartz-in-Garnet Barometry of an Ancient Subduction Zone; Cedros Island, Baja California Sur, Mexico. *Journal of Metamorphic Geology*. In review.
2025	**Xu, Y.**, Holder, R. M., Viete, D. R. (2025). Contrasting cooling histories in the Archean Minnesota River Valley Subprovince of the Superior Craton of North America with implications for secular change of metamorphic time scales. *GSA Bulletin,* 137(9-10): 3941-3955.
2024	**Xu, Y.**, Miller, C. F., Claiborne, L. L. (2024). Estimating Ce^{4+} and Ce^{3+} concentrations in zircon and coexisting melts, with implications for constraining oxygen fugacity. *Chemical Geology,* 663: 122283.
2021	de Wet, C., Erhardt, A., Sharp, W., Marks, N., Bradbury, H., Turchyn, A., **Xu, Y.**, Oster, J. (2021). Semi-quantitative estimates of rainfall variability during the 8.2kyr event in California using speleothem calcium isotope ratios. *Geophysical Research Letters,* 48(3).

CONFERENCE PRESENTATIONS

2025	**Xu, Y.***, Holder, R. M., Neumann, S., Connop, C. Rapid exhumation of the subduction mélange of Cedros Island, Baja California, Mexico from compositional zoning of garnet. AGU 2025, New Orleans, LA, USA.
2025	**Xu, Y.***, Holder, R. M., Neumann, S., Connop, C. Garnet Diffusion Chronometry Reveals Ultrafast Exhumation of the Subduction Mélange of Cedros Island, Baja California, Mexico. (Poster presentation) GSA Connects 2025, San Antonio, TX, USA.
2025	Holder, R. M., **Xu, Y.***, Hames, W. Rapid cooling of the Neoarchean Quetico metamorphic belt challenges the paradigm of long-lived, slow-cooling Archean metamorphism. GSA Connects 2025, San Antonio, TX, USA.
2025	Connop, C. H.*, Holder, R. M., Neumann, S. R., **Xu, Y.**, Hernández-Uribe, D., Wang, J. W., Kapp, P., He, J. J. Y., Worthington, J. Early Metamorphism of High-Grade Blocks from the Subduction Mélange of Cedros Island, Baja California, Mexico: Insights from Garnet Lu-Hf Geochronology. GSA Connects 2025, San Antonio, TX, USA.

2025	Holder, R. M.*, Hernández-Uribe, D., Wang, J. W., Connop, C. H., Neumann, S. R., **Xu, Y.**, Kapp, P., He, J. J. Y., Worthington, J. Empirical constraints on the long-term evolution of subduction-zone thermal structure: P–T–t estimates from the Cedros Island subduction complex, Baja California. GSA Penrose Conference - International Eclogites Conference
2025	Connop, C. H.*, Holder, R. M., Neumann, S. R., **Xu, Y.**, Hernández-Uribe, D., Wang, J. W., Kapp, P., He, J. J. Y., Worthington, J. The age of enigmatic high-grade blocks in subduction mélange of Cedros Island and their relationship to the tectonic evolution of the North American Cordillera: Garnet Lu-Hf and Sm-Nd geochronology. GSA Penrose Conference - International Eclogites Conference
2024	**Xu, Y.***, Holder, R. M. Did Metamorphic Terranes Cool Slower in the Past? Investigating Cooling Rates from Garnet Diffusion Chronometry on the Neoarchean Quetico Subprovince, Canada. AGU Fall Meeting 2024, Washington D.C., USA.
2024	**Xu, Y.***, Holder, R. M.. Did Archean Metamorphic Terranes Cool Slower? Garnet Diffusion Study of the Quetico Subprovince, Canada. (Poster presentation) Granulites & Granulites, Verbania, Italy.
2024	**Xu, Y.***, Holder, R. M.. Cooling of an Archean metamorphic terrane: garnet diffusion study of the Quetico Subprovince, Canada. (Poster presentation) 70th Annual Institute on Lake Superior Geology, Houghton, MI, USA.
2023	**Xu, Y.***, Holder, R. M., Viete, D. R. Interrogating the record of Archean tectonics from metamorphism in the Minnesota River Valley, USA (Superior Province). 6th International Archean Symposium, Fremantle, Australia.
2023	**Xu, Y.***, Holder, R. M., Viete, D. R. Interrogating the record of Archean tectonics from metamorphism in the Minnesota River Valley, USA (Superior Province). GSA North-Central Section Meeting, Grand Rapids, MI, USA.
2022	**Xu, Y.***, Holder, R. M., Viete, D. R. Interrogating the Record of Archean Tectonics from Metamorphism in the Minnesota River Valley, USA (Poster presentation). AGU Fall Meeting 2022, Chicago, IL, USA.
2020	**Xu, Y.***, Claiborne, L. L., Miller, C. F. Evaluating the application of Ce-in-zircon as a magma fO_2 estimator (Poster presentation). GSA Connects 2020 Online.

RESEARCH EXPERIENCE

2026-present **Composition of solid materials in protoplanetary disks** University of Michigan

- Apply phase-equilibrium modeling to predict compositions of condensed solids
- Compute opacity of the condensed dust particles
- Collaborate with the Department of Astronomy to integrate physical and chemical models

2021-2026 **PhD dissertation: "The Temporal Footprint of Tectonics: Deciphering Metamorphic Timescales via Diffusion Chronometry"** University of Michigan

- Investigated metamorphism in the Archean Superior Province using a variety of thermobarometers
- Applied monazite U–Pb geochronology with LA-ICP-MS and biotite Ar–Ar thermochronology
- Developed a Python code for modeling major element diffusion in garnet to constrain metamorphic cooling rates.
- Assessed the global time series of metamorphic timescales and its implications for secular change of tectonic processes.
- Identified rapid, pulsed subduction and exhumation of the mélange of Cedros Island, Baja California.

2025-present **Developing calcite reference material for U–Pb dating** University of Michigan

- Characterize the spatial heterogeneity of a calcite crystal with EPMA and LA-ICP-MS

2025-present **Metamorphic and structural history of Rainy Lake area,** University of Michigan

Minnesota, US.

- Field structural observations for relative timing of deformation and metamorphism.
- RAMAN spectrometry on quartz inclusions in garnet for pressure estimation.

2018-2021 **Undergraduate Honors Thesis: zircon trace element geochemistry** Vanderbilt University

- Applied zircon U–Pb geochronology and geochemistry to understand the paleotectonic origin of the Ordovician bentonite deposits in Tennessee.
- Compiled and analyzed zircon trace element data to evaluate the application of Ce-in-zircon as a proxy for magma oxygen fugacity.

2019 **Undergraduate Research Assistant** Vanderbilt University

- Collected water, soil, and calcite samples from two Karst caves in CA for isotopic analysis.
- Analyzed speleothem calcium and strontium isotopes with ICP-MS and investigated them as proxies for paleorainfall.

TEACHING EXPERIENCE

2022-2025 **Graduate student instructor** University of Michigan

- Courses taught: Field Geology, Structural Geology, Geochronology, Introductory Geology, Earth Math, Data and Models, Petrology
- Received U-M Graduate Teacher Certificate

2020 **Undergraduate student tutor** Vanderbilt University

- Led weekly tutor sessions on Dynamic Earth for student athletes.

GRANTS, AWARDS AND HONORS

2025 **Lewis and Clark Fund for Exploration and Field Research**, American Philosophical Society

2024 **Turner Student Research Award**, with Gold Star Award for best proposals, EARTH, University of Michigan

2024 **Best Graduate Poster**, 70th Institute on Lake Superior Geology

2022 **GSA Graduate Student Research Grant**, Geological Society of America

2021-2022 **Turner Student Research Award**, EARTH, University of Michigan

2019-2020 **Vanderbilt Undergraduate Summer Research Program Fellowship**, Vanderbilt University

2017-2020 **Dean's List Award**, College of Arts and Sciences, Vanderbilt University

TECHNICAL SKILLS

Sample preparation and mineral separation.
SEM. EPMA. LA-ICP-MS. Raman spectroscopy.
Phase-equilibrium modeling software: Theriak-Domino. MAGEMin. FastChem.
Python. ArcGIS. QGIS.

OUTREACH, MENTORSHIP AND COMMITTEE WORK

2025 **Co-convener,** AGU25 session V009 "Early Earth Origin, Composition, and Evolution" AGU

2024 **Graduate Student Mentor** EARTH, University of Michigan

2024 **Volunteer**, University of Michigan Museum of Natural History ID Day University of Michigan

2024 **Organizer committee**, 20th Annual Michigan Geophysical Union Symposium EARTH, University of Michigan

2022	**Unlearning Racism in Geoscience (URGE)**	EARTH, University of Michigan
2020	**Reviewer**, Vanderbilt Undergraduate Research Journal	Vanderbilt University

FIELD AND PROFESSIONAL EXPERIENCE

2026	Franciscan Complex, CA, USA
2025	Voyageurs National Park, MN, USA
2025	Short course: Teaching Structure Lab Using Math and Simple Computation, University of Wisconsin-Madison
2024	Ivrea Zone and the Central Alps
2024	Workshop "Phase Equilibrium Modelling with MAGEMin", Verbania, Italy
2023	Southern Ontario, Canada
2023	Field trip to the Yilgarn Craton, western Australia
2022	Quetico Subprovince, Ontario, Canada
2022	Metamorphism and structures in the Upper Peninsula of Michigan
2022	High-*T* metamorphism in the McCullough Range, NV, and Ivanpah Mtns, CA, USA
2022	Immersive Field Experience Exploring Earth Evolution (field course), Nevada, USA
2019	Speleothem sampling from caves in California, USA